TASK 5: Write Correct Expressions with the Right Precedence

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1. Aim

To apply operator precedence and associativity correctly in fare calculation and understand increment, division, remainder, and conditional operators.

2. Important Expressions

Peak-hour condition:

(hour >= 8 && hour < 10) ||

(hour >= 17 && hour < 20)

Peak surcharge:

farePaise = farePaise * 120 / 100;

Conditional operator:

long long fare = peak ? baseFare * 120 / 100 : baseFare;

The assignment does not specify the student or senior-citizen concession percentage, so that percentage must be confirmed before adding a concession expression.

3. C++ Program

```
#include <iostream>

using namespace std;

int main() {

    int x = 5, y = 3;

 cout << x + y * 2 << ' ' << (x + y) * 2 << '\n';
```

```
int z = x++ + ++y;
cout << z << ' ' << x << ' ' << y << '\n';
cout << (x > y ? x : y) << ' ' << x % 4 * 2 << '\n';
cout << 10 / 3 << ' ' << 10 % 3 << ' '
     << -10 / 3 << ' ' << -10 % 3 << '\n';
int k = 0;
for (int i = 0; i < 10; i += 3)
    k += i;
cout << k << '\n';}
```

4. Predicted Output

11 16

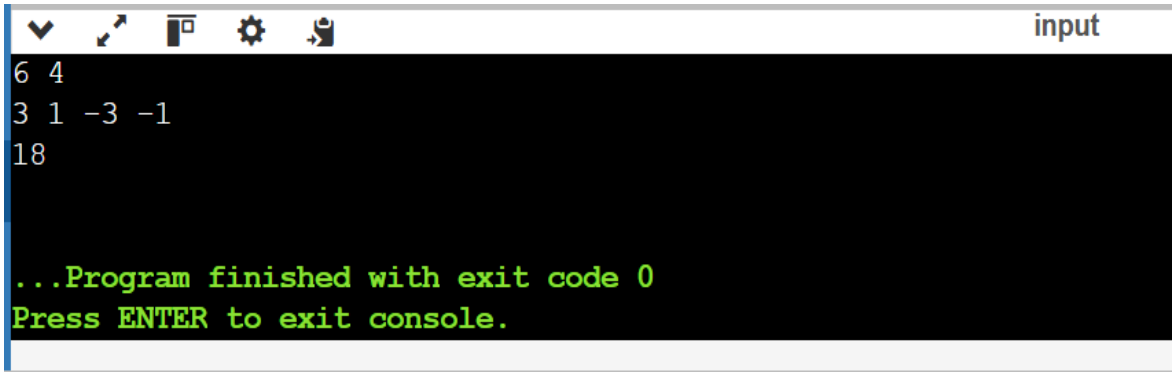
9 6 4

6 4

3 1 -3 -1

18

Obtained output :



5. Explanation

- x + y * 2: multiplication occurs first, giving 11.
- (x + y) * 2: parentheses force addition first, giving 16.
- x++ uses the old value of x and then increments it. ++y increments y before using it. Therefore, z = 5 + 4 = 9.
- After the increment expression, x = 6 and y = 4.
- x > y ? x : y returns the larger value, 6.
- % and * have the same precedence and associate left to right, so 6 % 4 * 2 gives 4.
- Integer division truncates toward zero: -10 / 3 gives -3, and the remainder is -1.
- The loop adds 0 + 3 + 6 + 9 = 18.

6. Increment Operators

- x++: returns the original value, then increments.
- ++x: increments first, then returns the new value.
- Avoid expressions such as i = i++; they are confusing and may have undefined behaviour in some language versions or expression contexts. Use i++ or ++i as a separate statement.

7. Alternative Concept and Complexity

A lookup table bool peak[24] can be used to store whether each hour is peak. Both the Boolean expression and the table lookup take **O(1) time**. The table uses 24 Boolean entries, with the exact byte size depending on the implementation.

8. Conclusion

Parentheses improve readability, and understanding precedence prevents incorrect fare calculations. Increment and remainder operators must be used carefully, especially in expressions with multiple operations.